VISTA User Guide

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Digital Business Valuation Midterm
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1. Access and first check

Open https://vista-ariel-dewa.streamlit.app/. No login is required. Public hosting requires internet and may sleep when inactive. If a wake-up button appears, click it and wait. Open the sidebar with the upper-left control on a narrow screen.

The submitted case uses a 19 October 2026 market reference and a 30 September 2026 financial position. It has no live market feed. Leave Scenario on Base, price at 242, WACC at 14%, terminal growth at 3.5% and terminal ROIC at 15%.

Expected starting result	Value
Blended / DCF / relative value per share	Rp152.3 / Rp79.8 / Rp209.1
Confidence / valuation gap	34/100 / -37.1%
Recommendation	PROVISIONAL AVOID / REDUCE

2. Run the submitted source locally

Extract VISTA_Project.zip. Install Python 3.12 if it is not already available. Open a terminal in the extracted project folder and run the following commands. On macOS/Linux the Python command may be python3.

```
python -m pip install -r requirements.txt
python -m streamlit run app.py
```

Open the local URL printed by Streamlit and keep the terminal open. Alternatively, run **python launch.py**; it creates a project virtual environment and installs the pinned packages when needed. The Mac shortcut Launch_VISTA.command invokes this launcher.

Initial installation needs internet. Streamlit and pandas are the required application packages. No paid data subscription, API key or AI service is needed to run the calculation engine. Localhost links only work on the computer running the application.

Required files: app.py, engine.py, model_io.py, requirements.txt and data/model_inputs.json. Keep the supplied PDFs and workbook in the same project folder to enable their downloads. The optional legacy web_app.py interface uses the same engine but does not provide the new structured-import controls.

Inputs, calculations and outputs

3. Change the operating case

Use the sidebar to choose Bear, Base or Bull, set the reference price and share count, or change WACC, terminal growth, terminal ROIC, tax, asset recoveries and quality judgments. The main result recalculates when an input changes.

Expand **Edit five-year operating assumptions** to change revenue growth, EBIT margin, D&A, cash CapEx and NWC ratios. Entries in this table are percentages: enter 8 for 8%. Commit a cell with Enter or by moving to another cell.

Expand **Peer selection and multiple overrides** to include/exclude peers and change EV/EBITDA. Included multiples must be positive and finite. Removing every peer makes relative valuation unavailable and assigns the blend to DCF.

4. Calculation logic

Stage	Logic
Normalization	Revenue less cost of sales, SG&A; and final tax gives recurring EBIT. Acquisition gains do not enter recurring operating profit.
FCFF DCF	EBIT after normalized income tax + D&A; - cash and lease CapEx - increase in NWC. Discount five years at WACC and add terminal value.
Terminal value	Next-year NOPAT x (1 - g / ROIC) / (WACC - g). This explicitly funds growth.
Relative valuation	Median included peer EV/EBITDA x normalized annualized operating EBITDA.
Equity and recommendation	Add cash and asset recoveries, deduct debt and NCI, divide by shares. Suitability scores weight the methods. Compare the blend with price.

5. Navigate and interpret

Valuation: forecast and equity bridge. **Financial history:** cumulative periods and submitted 2026 quarterly derivation. **Scenarios & sensitivity:** saved cases and WACC/g grid. **Suitability & confidence:** weights and simulation. **Sources & checks:** reconciliations, sources and limitations.

Named scenarios retain their saved operating drivers; active market, peer and quality inputs apply to all. The edited scenario appears in the main dashboard. The simulation starts from the saved Base case, including an imported Base, rather than active slider overrides.

A gap above +20% signals BUY CANDIDATE; below -20% signals AVOID / REDUCE; otherwise WATCH / HOLD. Confidence below 60 adds PROVISIONAL. Neither the score nor scenario probabilities are measured probabilities that the valuation is correct.

Updating the model without rebuilding it

6. Structured import workflow

1. Expand **Update financial and market data** and download the model update template. Keep an untouched backup. Open the JSON in a plain-text or JSON editor.
2. Update the relevant fields below, their sources and dates. Preserve the field names and prior periods.
3. Select the edited file under **Import updated model JSON** and click **Validate and apply import**.
4. Review the success/error message, data dates, Sources & Checks and all three scenarios. A rejected import leaves the current dataset unchanged.
5. Use **Save active model inputs** to keep the session changes and re-import them later.

JSON area	What to update
historical	Cumulative income/cash-flow figures and period-end balance sheet. Values are IDR billion. Add a complete snapshot for a new period using the existing latest-period fields.
latest_period / latest_period_months	Set the new historical key and cumulative months (3, 6, 9 or 12). Calendar-year statements only. Annualization is 12 / months.
financial_date / valuation_date / prepared	Use YYYY-MM-DD. Financial date must match the cumulative month and cannot follow the market valuation date. Update prepared to the input revision date.
market / peers	Price, shares_m (millions), source URLs, dates, peer multiples and selection. Harmonize price and earnings endpoints where possible.
scenarios / assumptions	Keep Bear, Base and Bull. Ratios in JSON use decimals: 0.08 means 8%. Probabilities sum to one.
records / exploration_tax_period	Update source file, physical page, units and values. Preserve FY2025 acquisition evidence. Select the historical period containing the exploration_dtl used in the recovery bridge.

7. Reconciliation and persistence

For a new cumulative cash-flow period, supply opening_cash and reconcile it with CFO, CFI, CFF and closing cash. Update all affected income and balance-sheet components together. The importer checks arithmetic, not whether an external source is true or a financial item is correctly classified.

Imports stay in the current session and do not change GitHub. To publish a revised default, replace data/model_inputs.json in the repository, retain its citations and archive supporting filings. Update source_inputs.csv, the workbook and narrative outputs separately. The fixed 2026 explanatory notes and quarter table must also be reviewed after a period change.

Export, validation and limitations

8. Save results and use the workbook

In Scenarios & sensitivity, download the active forecast CSV, the active valuation JSON (with export timestamp and dates), or the active input JSON. Fixed downloads contain the submitted Excel, report, slide PDF and this guide. Session changes do not automatically rewrite those documents.

In Excel, choose the case in Controls D4, edit blue input cells and inspect the source register and Audit sheet. Use Automatic calculation, including data tables. The workbook is formula-based .xlsx without VBA; the Python application is the primary functional deliverable.

9. Validation and a quick acceptance test

From the project folder, run **python -m unittest -v test_engine test_import**. The submission passes 24 engine/import tests. The original workbook checks agree with Python for three scenarios and all 25 sensitivity cells. Native desktop Excel was not automated.

Try WACC 14% to 16%: Base DCF should fall from Rp79.8 to about Rp54.1. Reset, then change price 242 to 200: fair value remains Rp152.3 while the gap changes to about -23.8%. Download and re-import an unchanged template: results remain the same. A wrong unit or unreconciled balance sheet should be rejected.

10. Known limitations and troubleshooting

Exact treasury-adjusted diluted shares, harmonized peer earnings, equipment/lease commitments and mine economics remain unverified. WACC, growth and recovery rates are assumptions. Nine-month annualization can miss seasonality. Terminal PV is 72% of Base DCF operating EV. The scenario range is conditional, and simulation percentiles describe selected assumptions.

If the public app is unavailable, run the local package. If an import fails, read the message and correct the named field/reconciliation. Use Restore submitted data to reset. If a document is absent, the app shows an availability message instead of crashing. Browser locale may affect editable input display; tables use comma thousands and dot decimals.

11. AI disclosure and responsibility

OpenAI ChatGPT and Codex assisted extraction, coding, formulas, analysis and document design. Checks included source-page review, financial reconciliations, independent FCFF tests, Excel/Python comparison and browser testing. Corrections included a Q1 prior-year column reading and a missing-PPTX download error. These checks do not independently certify the source data or investment conclusion.

Ariel Arviano Putra remains responsible for the submitted model, assumptions and conclusions, and for explaining the work independently. Full disclosures appear in the slide report and methodology addendum.